Year 10 Higher

REVISION PACK

Progress Check 3

ANSWERS

Units in this pack

- Number 3
- Algebra 2

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Number 3 · Negative and fractional indices (Lessons 6.1, 6.2)

Q	Working	Answer
working out the value of powers		
1.1	**(a)** $3^{-2} = \frac{1}{3^2}$ **(b)** $\left(\sqrt[3]{27}\right)^2$ $= 3^2 = 9$ **(c)** $\left(\frac{27}{64}\right)^{\frac{1}{3}}$ $= \frac{3}{4}$	**(a)** $\frac{1}{9}$ **(b) 9** **(c)** $\frac{3}{4}$
1.2	**(a)** $2^{-5} = \frac{1}{2^5}$ **(b)** $\left(\sqrt{16}\right)^3$ $= 4^3 = 64$ **(c)** $\left(\frac{16}{9}\right)^{\frac{1}{2}}$ $= \frac{4}{3}$	**(a)** $\frac{1}{32}$ **(b) 64** **(c)** $\frac{4}{3}$
1.3	**(a)** $10^{-3} = \frac{1}{10^3}$ **(b)** $\left(\sqrt[3]{125}\right)^2$ $= 5^2 = 25$ **(c)** $\left(\frac{125}{8}\right)^{\frac{1}{3}}$ $= \frac{5}{2}$	**(a)** $\frac{1}{1000}$ **(b) 25** **(c)** $\frac{5}{2}$

1.4	**(a)** $4^{-2} = \frac{1}{4^2}$ **(b)** $\left(\sqrt{9}\right)^3$ $= 3^3 = 27$ **(c)** $\left(\frac{36}{25}\right)^{\frac{1}{2}}$ $= \frac{6}{5}$	**(a)** $\frac{1}{16}$ **(b)** **27** **(c)** $\frac{6}{5}$
1.5	**(a)** $5^{-3} = \frac{1}{5^3}$ **(b)** $\left(\sqrt[3]{8}\right)^4$ $= 2^4 = 16$ **(c)** $\left(\frac{64}{27}\right)^{\frac{1}{3}}$ $= \frac{4}{3}$	**(a)** $\frac{1}{125}$ **(b)** **16** **(c)** $\frac{4}{3}$
1.6	**(a)** $2^{-6} = \frac{1}{2^6}$ **(b)** $\left(\sqrt{100}\right)^3$ $= 10^3 = 1000$ **(c)** $\left(\frac{4}{49}\right)^{\frac{1}{2}}$ $= \frac{2}{7}$	**(a)** $\frac{1}{64}$ **(b)** **1000** **(c)** $\frac{2}{7}$
finding the power n		
1.7	**(a)** $\frac{1}{16} = \frac{1}{2^4} = 2^{-4}$ **(b)** $n - 3 = -1$	**(a)** $n = -4$ **(b)** $n = 2$
1.8	**(a)** $\frac{1}{27} = \frac{1}{3^3} = 3^{-3}$ **(b)** $n - 5 = -2$	**(a)** $n = -3$ **(b)** $n = 3$

1.9	**(a)** $\frac{1}{100} = \frac{1}{10^2} = 10^{-2}$ **(b)** $n - 7 = -3$	**(a)** $n = -2$ **(b)** $n = 4$
1.10	**(a)** $\frac{1}{64} = \frac{1}{4^3} = 4^{-3}$ **(b)** $n - 9 = -4$	**(a)** $n = -3$ **(b)** $n = 5$
1.11	**(a)** $\frac{1}{81} = \frac{1}{9^2} = 9^{-2}$ **(b)** $n - 3 = -2$	**(a)** $n = -2$ **(b)** $n = 1$
1.12	**(a)** $\frac{1}{8} = \frac{1}{2^3} = 2^{-3}$ **(b)** $n - 6 = -1$	**(a)** $n = -3$ **(b)** $n = 5$

Algebra 2 · Index laws with algebra (Lesson 7.1)

Q	Working	Answer
simplifying with index laws		
2.1	**(a)** $2^3 = 8,\ (x^4)^3 = x^{12},\ (y^2)^3 = y^6$ **(b)** $18 \div 6 = 3,\ x^5,\ y^{-2}$	**(a)** $8x^{12}y^6$ **(b)** $\frac{3x^5}{y^2}$
2.2	**(a)** $5^2 = 25,\ a^4,\ b^6$ **(b)** $20 \div 4 = 5,\ a^3,\ b^{-2}$	**(a)** $25a^4b^6$ **(b)** $\frac{5a^3}{b^2}$
2.3	**(a)** $3^4 = 81,\ x^{20},\ y^8$ **(b)** $24 \div 8 = 3,\ m^5,\ n^{-4}$	**(a)** $81x^{20}y^8$ **(b)** $\frac{3m^5}{n^4}$
2.4	**(a)** $2^3 = 8,\ a^9,\ b^{12}$ **(b)** $30 \div 6 = 5,\ a^2,\ b^{-4}$	**(a)** $8a^9b^{12}$ **(b)** $\frac{5a^2}{b^4}$
2.5	**(a)** $4^3 = 64,\ x^6,\ y^3$ **(b)** $21 \div 7 = 3,\ x^3,\ y^{-3}$	**(a)** $64x^6y^3$ **(b)** $\frac{3x^3}{y^3}$

2.6	**(a)** $2^4 = 16,\ m^{20},\ n^8$ **(b)** $16 \div 4 = 4,\ p^4,\ q^{-3}$	**(a)** $16m^{20}n^8$ **(b)** $\dfrac{4p^4}{q^3}$
fractional powers with letters		
2.7	$(\sqrt[3]{125})^2 = 5^2 = 25$ $6 \times \frac{2}{3} = 4, \quad 9 \times \frac{2}{3} = 6$	$25x^4y^6$
2.8	$(\sqrt[3]{64})^2 = 4^2 = 16$ $9 \times \frac{2}{3} = 6, \quad 3 \times \frac{2}{3} = 2$	$16x^6y^2$
2.9	$(\sqrt{25})^3 = 5^3 = 125$ $4 \times \frac{3}{2} = 6, \quad 8 \times \frac{3}{2} = 12$	$125x^6y^{12}$
2.10	$(\sqrt[3]{8})^2 = 2^2 = 4$ $6 \times \frac{2}{3} = 4, \quad 12 \times \frac{2}{3} = 8$	$4x^4y^8$
2.11	$(\sqrt[4]{16})^3 = 2^3 = 8$ $8 \times \frac{3}{4} = 6, \quad 12 \times \frac{3}{4} = 9$	$8x^6y^9$
2.12	$(\sqrt[3]{1000})^2 = 10^2 = 100$ $3 \times \frac{2}{3} = 2, \quad 6 \times \frac{2}{3} = 4$	$100x^2y^4$

Algebra 2 · Algebraic fractions (Lesson 7.2)

Q	Working	Answer
simplifying algebraic fractions		
3.1	$\dfrac{x(x-5)}{(x+5)(x-5)}$	$\dfrac{x}{x+5}$
3.2	$\dfrac{x(x+2)}{(x+2)(x-2)}$	$\dfrac{x}{x-2}$
3.3	$\dfrac{(x+6)(x-6)}{3x(x+6)}$	$\dfrac{x-6}{3x}$
3.4	$\dfrac{x(x-7)}{(x+7)(x-7)}$	$\dfrac{x}{x+7}$
3.5	$\dfrac{x(x+6)}{(x+6)(x-6)}$	$\dfrac{x}{x-6}$

3.6	$\dfrac{(x+1)(x-1)}{2x(x+1)}$	$\dfrac{x-1}{2x}$
fractions with a quadratic		
3.7	$\dfrac{(x+5)(x-2)}{(x+2)(x-2)}$	$\dfrac{x+5}{x+2}$
3.8	$\dfrac{(x-4)(x+3)}{(x+4)(x-4)}$	$\dfrac{x+3}{x+4}$
3.9	$\dfrac{(x+2)(x+3)}{(x+3)(x-3)}$	$\dfrac{x+2}{x-3}$
3.10	$\dfrac{(x+5)(x-5)}{(x-5)(x+2)}$	$\dfrac{x+5}{x+2}$
3.11	$\dfrac{(x+5)(x-1)}{(x+1)(x-1)}$	$\dfrac{x+5}{x+1}$
3.12	$\dfrac{(x-4)(x+2)}{x(x-4)}$	$\dfrac{x+2}{x}$
dividing algebraic fractions		
3.13	$\dfrac{x+3}{5} \times \dfrac{10}{x^2-9}$ $= \dfrac{x+3}{5} \times \dfrac{10}{(x+3)(x-3)}$	$\dfrac{2}{x-3}$
3.14	$\dfrac{x-4}{2} \times \dfrac{6}{x^2-16}$ $= \dfrac{x-4}{2} \times \dfrac{6}{(x+4)(x-4)}$	$\dfrac{3}{x+4}$
3.15	$\dfrac{x+1}{6} \times \dfrac{3}{x^2-1}$ $= \dfrac{x+1}{6} \times \dfrac{3}{(x+1)(x-1)}$	$\dfrac{1}{2(x-1)}$
3.16	$\dfrac{x+6}{4} \times \dfrac{12}{x^2+6x}$ $= \dfrac{x+6}{4} \times \dfrac{12}{x(x+6)}$	$\dfrac{3}{x}$
3.17	$\dfrac{x-7}{3} \times \dfrac{9}{x^2-49}$ $= \dfrac{x-7}{3} \times \dfrac{9}{(x+7)(x-7)}$	$\dfrac{3}{x+7}$

3.18	$\frac{x^2-4}{5} \times \frac{10}{x+2}$ $= \frac{(x+2)(x-2)}{5} \times \frac{10}{x+2}$	$2(x-2)$

Algebra 2 · Equations with powers and roots (Lesson 7.3)

Q	Working	Answer
solving equations with roots and squares		
4.1	**(a)** $2x+5=9$ $2x=4$ $x=2$ **(b)** $2x^2=50$ $x^2=25$ $x=\pm5$	**(a)** $x=2$ **(b)** $x=5$ **or** $x=-5$
4.2	**(a)** $4x+1=25$ $4x=24$ $x=6$ **(b)** $4x^2=36$ $x^2=9$ $x=\pm3$	**(a)** $x=6$ **(b)** $x=3$ **or** $x=-3$
4.3	**(a)** $5x-4=36$ $5x=40$ $x=8$ **(b)** $3x^2=27$ $x^2=9$ $x=\pm3$	**(a)** $x=8$ **(b)** $x=3$ **or** $x=-3$
4.4	**(a)** $2x+1=25$ $2x=24$ $x=12$ **(b)** $5x^2=80$ $x^2=16$ $x=\pm4$	**(a)** $x=12$ **(b)** $x=4$ **or** $x=-4$

4.5	**(a)** $6x - 2 = 16$ $6x = 18$ $x = 3$ **(b)** $2x^2 = 50$ $x^2 = 25$ $x = \pm 5$	**(a)** $x = 3$ **(b)** $x = 5$ **or** $x = -5$
4.6	**(a)** $3x - 5 = 4$ $3x = 9$ $x = 3$ **(b)** $4x^2 = 36$ $x^2 = 9$ $x = \pm 3$	**(a)** $x = 3$ **(b)** $x = 3$ **or** $x = -3$
solving with a bracket squared		
4.7	$x + 2 = \pm 7$ $x = 5$ or $x = -9$	$x = 5$ **or** $x = -9$
4.8	$x - 5 = \pm 6$ $x = 11$ or $x = -1$	$x = 11$ **or** $x = -1$
4.9	$x + 6 = \pm 4$ $x = -2$ or $x = -10$	$x = -2$ **or** $x = -10$
4.10	$x - 1 = \pm 8$ $x = 9$ or $x = -7$	$x = 9$ **or** $x = -7$
4.11	$x + 3 = \pm 10$ $x = 7$ or $x = -13$	$x = 7$ **or** $x = -13$
4.12	$x - 7 = \pm 3$ $x = 10$ or $x = 4$	$x = 10$ **or** $x = 4$
cube roots and square roots		
4.13	$e^3 = 150$ $e = \sqrt[3]{150} = 5.3133\ldots$	**5.31 cm**

4.14	$s^2 = 90$ $s = \sqrt{90} = 9.4868\ldots$	**9.49 cm**
4.15	$e^3 = 500$ $e = \sqrt[3]{500} = 7.9370\ldots$	**7.94 cm**
4.16	$s^2 = 45$ $s = \sqrt{45} = 6.7082\ldots$	**6.71 cm**
4.17	$e^3 = 80$ $e = \sqrt[3]{80} = 4.3089\ldots$	**4.31 cm**
4.18	$e^3 = 1200$ $e = \sqrt[3]{1200} = 10.6266\ldots$	**10.6 cm**